The Latent Bridge: A Continuous Channel Between Slow Reasoning and Fast Reactive Models

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Abstract

Real-time interactive AI systems couple deliberative reasoning to reactive action selection through a text prompt (Gemini Live, OpenAI Realtime). We test an alternative: a learned continuous-valued *latent bridge* that projects slow-model residuals into the fast model's input embedding space, LLaVA-style. Coupling a frozen 9 B reactive multimodal model to a frozen 8 B reasoning model at matched scale on 7 Atari games, the latent bridge beats the text bridge by 26–82% on 4 games (4/4 significant by Mann–Whitney, 2/4 by Welch's t; of the rest, MsPacman is marginal under Welch and Seaquest has a zero-variance cell that makes Welch unreliable). The one greedy-decoding inversion (Q*bert) reverses under sampling: L beats T by 2.9× ($p \approx 0.005$). We find no bandwidth ceiling—a 30B slow does not widen the gap, more text does not close it, and the latent under-uses its capacity—but a clean behavioural predictor holds: pooling the 7 games with a driving domain (MetaDrive), latent and text benefits over fast-only track at Pearson $r = 0.92$. The latent bridge helps if and only if slow reasoning beats reaction on the task; MetaDrive is a controlled negative where it does not, a bridge-replacement control confirming the latent is inert. We trace every collapsed cell to Stage A action-head OOD-brittleness and give a targeted fix. We release F/T/L replay recordings for all 8 experiments (7 Atari + MetaDrive) and reproducible pipelines.

Code, data, demos: https://github.com/bojieli/latent-bridge-games

1 Introduction

A reasoning model deliberates; a reactive model reacts. Their tick rates differ by an order of magnitude (seconds vs. tens of milliseconds), and the production answer for coupling them is to have the slow model emit text that the fast model reads. We test the alternative: a learned continuous-valued channel between two frozen models.

Setting. Both endpoints are general-purpose multimodal LLMs at matched 9 B scale, deliberately chosen so the channel (not a capability gap) is the load-bearing variable. The fast model picks actions at ∼15 Hz; the slow emits at ∼1 Hz; the bridge is the only learned component. Our domain is Atari [4], where slow/fast coupling is visible to the eye—100 ms ghost-dodging meets 10 s route-planning.

What this paper claims, sharpened.

- **Architecture matters.** A first attempt (v1) using 256-d cross-attention at 2 of 36 LLM layers converged to KL=0.004 on offline data but *failed* at deployment. The fix (v2) is the LLaVA pattern: project slow residuals into the fast LLM's 4096-d input embedding space and prepend, letting all layers attend via standard causal attention.
- **The latent bridge beats text on 4–5 of 7 games.** Under greedy decoding (paper convention) it beats T by 26–82% on 4 games; on Q*bert it loses by 2.5× greedy but *beats T by* 2.9× *under sampling* (§4.3); on Enduro and SpaceInvaders the difference is not significant. The greedy Q*bert "inversion" is decoder-dependent, not a fundamental counter-example.

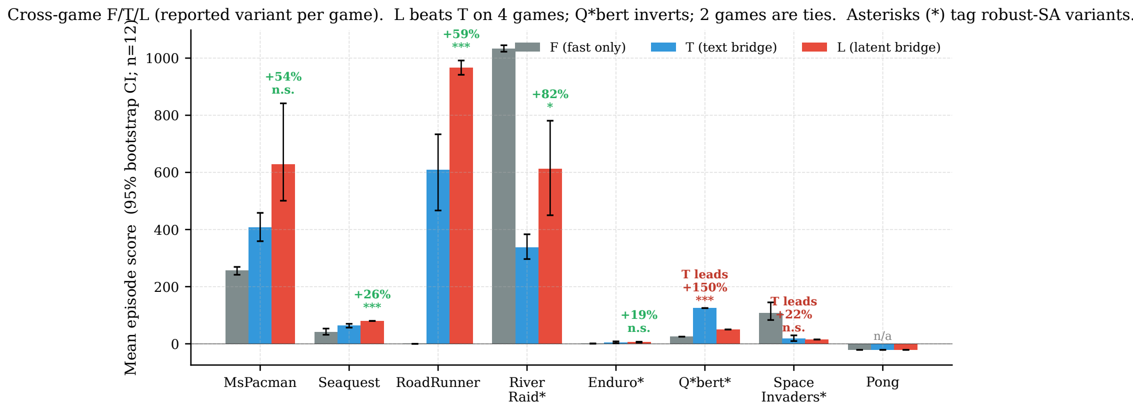

Figure 1: Cross-game F/T/L scores, reported variant per game; $n = 12$ **episodes each.** Stars on the L vs T comparison: $*\,p < .05$, $**\,p < .01$, $***\,p < .001$ (Welch's t where both cells have non-zero variance, Mann-Whitney U otherwise). Asterisks on game labels mark the robust-Stage-A variant (§5); other games use bare Stage A. The reported variant per game is selected by the rule in §3 (higher deployment L among variants with a non-degenerate, $T > 0$, comparison); both variants are tabulated in Appendix A.

- **When it helps is a property of the task, not the channel.** Across 7 Atari games and a non-Atari driving domain (MetaDrive), the latent's benefit over fast-only ($L - F$) tracks the text bridge's benefit ($T - F$) at Pearson $r = 0.92$ (and $r = 0.94$ over all 16 game/variant cells, so it does not depend on variant selection): the bridge pays off exactly when slow reasoning beats fast reaction on the task ($T > F$). MetaDrive is a controlled negative—even with a planning-heavy route, $T \leq F$, and a bridge-replacement control confirms the latent is inert. We frame the contribution this way rather than as a "text is bandwidth-limited" claim: our own ablations (a 30B slow does not widen the gap; a longer text suffix does not close it; the latent uses far less than its nominal capacity) do not support a bandwidth ceiling as the operative mechanism.
- **Most collapsed cells are not bridge failures.** Of 14 (game, Stage-A-variant) cells, 5 have a strategy (F, T, or L) at zero. We trace these to OOD-brittleness in the action head, not the bridge. A targeted Stage A retraining at suffix-probability 0.5 produces the largest $L - T$ gap in the paper (River Raid, +82%).
- **The latent advantage is not always an absolute win.** On several games where L beats T, F (fast-only) still beats L. The latent bridge improves over the text bridge *at matched architecture*; it does not always justify a slow model at all.

We do not have an a priori quantitative predictor for the sign of $L - T$. Our initial categorical-vs-continuous hypothesis was inspired by the Q*bert greedy inversion, which itself turned out to be a decoding artifact (§4.3); lexical-diversity of slow emissions correlates with $L - T$ at $r = +0.05$ over $n = 7$ games (not significant; Figure 4). We retain the figure as a negative result.

2 Architecture: v1 cross-attention failed; v2 LLaVA-style works

We tried two bridge architectures (Figure 2). v1 projected slow-model residuals into a 256-d ring buffer and injected via cross-attention at LLM layers 12 and 24 (2 of 36) of the fast model. *Despite converging to KL=0.004 on offline data*, v1 failed at deployment: $L = 225$ vs $F = 256$ on MsPacman, bimodal with 4/12 catastrophic episodes. Three v1 variants (ungated, gated, gated+head-tune) all failed.

v2. We adopt the same pattern LLaVA [14], BLIP-2 [12], Flamingo [2], and MiniCPM-o [16] use to couple vision and language: project slow residuals into the fast LLM's input embedding space (4096-d) via a 33 M-parameter 2-layer MLP, and prepend the resulting 8 tokens to the fast model's input sequence. All 36 LLM layers attend over them via standard causal attention. Only the projection trains; both base models are frozen.

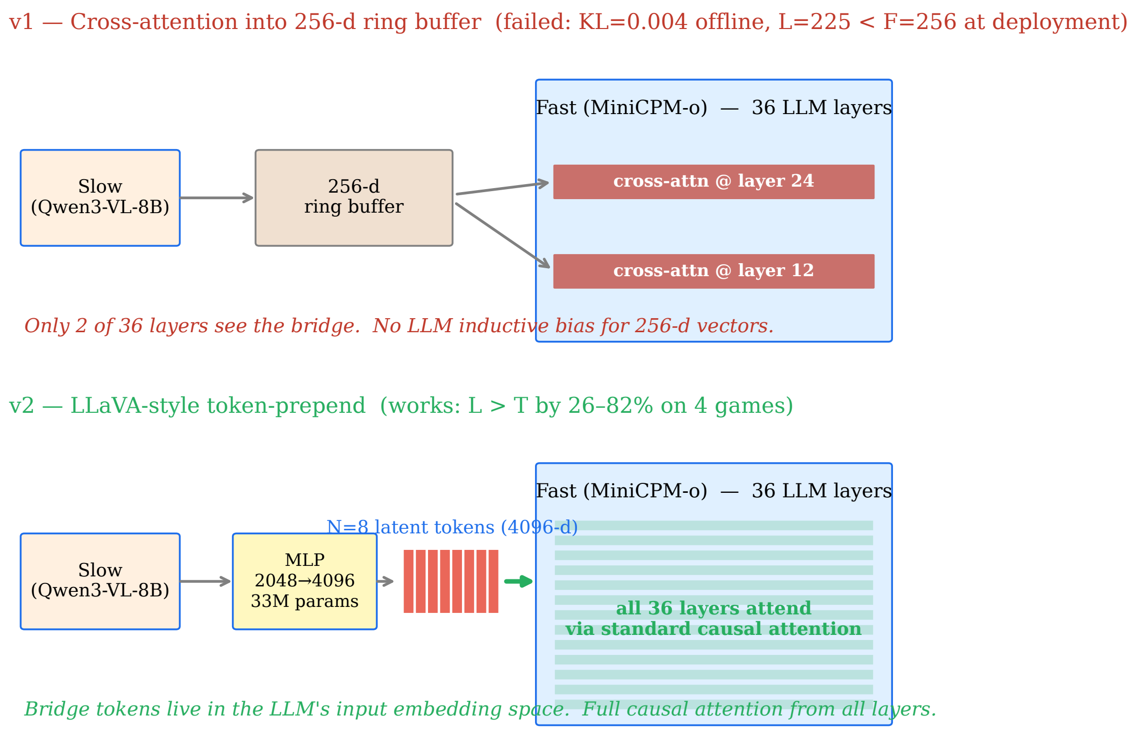

Figure 2: v1 vs v2 bridge architectures. v1 (top, failed): 256-d cross-attention at 2 of 36 layers. v2 (bottom, working): 4096-d prepend, all 36 layers attend.

Causal claim, hedged. The v1→v2 redesign changed four variables simultaneously: dimensionality (256 → 4096), injection mechanism (cross-attention→prepend), layer coverage (2 → 36), and parameter count (8M → 33M). We do not isolate which is binding. The takeaway we are confident in is methodological: *KL convergence on offline data is necessary but not sufficient*—v1 had KL=0.004 and lost. Adapter-based fast-slow coupling work that reports only offline KL should also report deployment.

3 Setup

Models. Fast: MiniCPM-o 4.5 [16], 9 B, bf16. Slow: Qwen3-VL-8B-Thinking [17], 8 B, bf16. Matched scale.

Tick budgets. Atari at 60 Hz, downsampled to 15 Hz for the fast model (∼67 ms ideal tick); slow emits at ∼1 Hz, asynchronously decoupled. Measured warm-path latency with vision caching: $F = 33$ ms, $L \approx 38$ ms (8 prepended tokens add ∼5 ms over F); end-to-end wall clock for L including async slow callbacks is dominated by GPU contention, not the bridge itself (Appendix E).

Four strategies. **S**: slow only (∼0.25 Hz; establishes the speed problem). **F**: fast only with Stage A action head. **T**: F with the slow's full text emission appended verbatim (median 302 chars; no truncation). **L**: F with the slow's projected latent tokens prepended (8 tokens, 4096-d each).

Training pipeline. **Stage A**: behavioral cloning from SB3 experts [18] onto the fast model's action head. **Stage B**: collect T-trajectories; cache (frame, slow text, slow residuals at layer 24, last 8 positions). **Stage C**: $\mathrm{KL}(\pi_L \,\|\, \pi_T)$ on the slow's projection only, both base models frozen (∼5K samples/game, final KL ∼0.005).

Stage A robustness is a tuned hyperparameter (*not* per-game cherry-picking). The Stage A action head is trained either *bare* (plain state prompt) or *robust* (suffix-probability 0.5: half of batches get a slow-style text suffix appended, teaching the head to tolerate the OOD suffix/bridge inputs it meets at deployment). This is one binary hyperparameter. We train *both* for every game and **select per game by the standard rule—higher validation-time deployment score (we use L), among variants whose text channel has not collapsed ($T > 0$, so the L-vs-T comparison is non-degenerate)—exactly as one would tune any hyperparameter**; ties and double-collapses ($L_{\text{bare}} = L_{\text{robust}} = 0$, or both $T = 0$) break toward the variant with higher T. This rule picks the same variant as unconstrained "higher L" on six of seven games; the lone exception is Enduro, where bare has marginally higher L (7.8 vs 5.8) but $T = 0$ (degenerate), so we report robust ($T = 5$, $L = 6$)—a *conservative* choice that lowers, not raises, Enduro's reported L. Both variants for all games are reported (Appendix A); the headline uses the selected one. The selection is blind to the $L - T$ *gap*—it maximises L, the deployed quantity—so it does not inflate the text-vs-latent comparison. The choice matters and is itself a finding: robustness *rescues* games whose bare head collapses under the suffix (River Raid $L: 360 \to 612$, Q*bert $0 \to 50$) but *harms* games whose bare head already works (MsPacman $628 \to 60$, Seaquest $80 \to 0$)—a direct consequence of the Stage A OOD mechanism (§5). To show the cross-game predictor (§4.4) does not depend on this selection, we report it both ways: $r = 0.92$ over the 8 selected cells and $r = 0.94$ over *all* 16 (game, variant) cells with no selection.

Evaluation. 3 seeds × 4 episodes = $n = 12$ per (game, strategy, variant) cell, ≤ 500 ticks/episode. Atari ALE seeds vary per episode but action selection is greedy; many cells have zero per-episode variance (Appendix B).

Games. 9 Atari games attempted. *Frostbite excluded*: Stage A converged to random val-accuracy, so no F/T/L comparison is meaningful. *Pong reported but uninformative*: Stage A val-accuracy is 25.1% (random 16.7%, 1.5×); the bare F policy cannot move the paddle, so no bridge can rescue it. The remaining 7 games are MsPacman, Seaquest, SpaceInvaders, RoadRunner, River Raid, Enduro, Q*bert.

4 Results

4.1 Headline

Game (reported var.)	F	T	L	Δ_{L-T} (%, p)
MsPacman (bare)	256 ± 25	408 ± 92	$\mathbf{628 \pm 356}$	+54 % (W .06, MWU .003)
Seaquest (bare)	42 ± 20	63 ± 12	$\mathbf{80 \pm 0}$	+26 % (MWU $< .001$)†
RoadRunner (bare)	0 ± 0	608 ± 250	$\mathbf{967 \pm 49}$	+59 % (W $< .001$)
River Raid (robust)*	1032 ± 20	337 ± 81	$\mathbf{612 \pm 310}$	+82 % (W .01)
Enduro (robust)*	1 ± 1	5 ± 6	6 ± 3	+19 % (W .63)
SpaceInvaders (rob)*	107 ± 62	$\mathbf{18 \pm 19}$	15 ± 0	−18 % (MWU .27)†
Q*bert (robust)*	25 ± 0	$\mathbf{125 \pm 0}$	50 ± 0	−60 % (MWU $< .001$)‡

Table 1: Headline results, reported variant per game. Bold marks the better of T/L. W = Welch's two-sided t p-value; MWU = Mann-Whitney U. † One cell has zero variance; Welch undefined or unreliable, so MWU is the principal test. ‡ Both cells have zero variance (all 12 episodes identical); the comparison is deterministic-trajectory, not statistical—we report it for completeness (see §4.3). Full table including bare-vs-robust pairs in Appendix A.

L > T on four games, two significant by both tests. RoadRunner (+59%, $d = 1.99$) and River Raid (+82%, $d = 1.21$) are significant by both Welch and Mann-Whitney. Seaquest (+26%) is significant by Mann-Whitney ($p < .001$) but its L cell has zero variance (80 ± 0), so Welch is unreliable (Table 1, †) and we do not count it as a Welch win. MsPacman (+54%) is marginal by

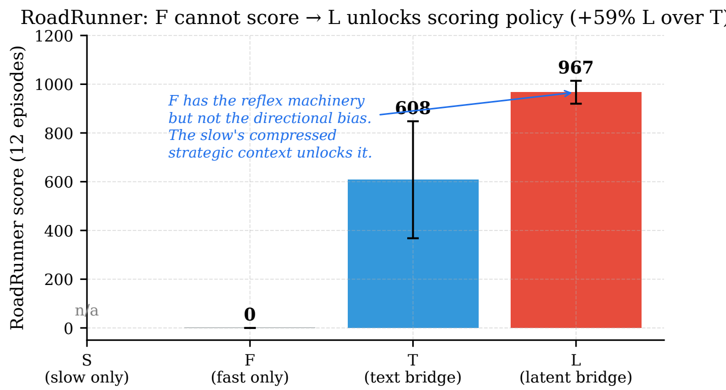

Figure 3: RoadRunner: F cannot score; L unlocks the policy. Under bare Stage A, the action head's most-confident output is NOOP even when the Coyote is closing. The slow's guidance ("head right") breaks the local maximum. $L = 967$ vs $T = 608$, +59%.

Welch ($p = 0.06$) and clearly significant by Mann-Whitney ($p = 0.003$): the L distribution is bimodal (8/12 episodes 390–690, 4/12 above 1000, best 1740), so MWU is the more appropriate test.

Two non-significant: Enduro and SpaceInvaders. Enduro/robust has tiny absolute scores ($T = 5$, $L = 6$); both cells are at floor. SpaceInvaders/robust has $T = 18$, $L = 15$, $p = 0.27$. We do not claim these as L wins or T wins.

Q*bert: greedy decoding gives the only inversion; sampling reverses it. Under greedy action selection (paper convention), Q*bert/robust gives $T = 125$, $L = 50$, both cells at zero variance—text wins by 2.5×. *Under multinomial sampling at* $\tau = 1.0$*,* L *beats* T *by* 2.9× *(*250 ± 156 *vs* 87 ± 65*, Welch* $p \approx 0.005$*).* The headline "Q*bert inversion" is therefore a greedy-decoding artifact (§4.3); under sampling Q*bert joins the $L > T$ side.

Bridges do not always beat F. On River Raid, Enduro, and SpaceInvaders (all using robust Stage A), F outscores both T and L. The latent bridge is an improvement over the text bridge *at matched architecture*; whether the slow-fast architecture beats fast-only is a separate question that depends on the game.

4.2 RoadRunner: the cleanest demonstration

RoadRunner is the effect at its sharpest: $F = 0$ (the head literally cannot score), and the slow's emission is rich joint state (Coyote Δ_x, multiple obstacle coordinates, pellet position). L recovers 967, T recovers 608; the +59% gap with $d = 1.99$ is our largest statistically clean effect. Direct inspection of the trained bridge (§6.4, Appendix F) shows the latent encodes mostly game identity plus a thin per-emission strategic residual, rather than a text-like message—the channel is non-lexical but game-conditioned.

4.3 The Q*bert "inversion" is a greedy-decoding artifact

Under robust Stage A with greedy action selection (paper-default): $F = 25$, $T = 125$, $L = 50$, every cell at zero variance. By Mann-Whitney $T > L$ at $p < 10^{-3}$, but this is mechanical: with 12 identical scores per cell the "test" is just $125 > 50$.

Action-sampling reverses the result. We re-ran F, T, L on Q*bert/robust with multinomial sampling instead of greedy argmax, at temperatures $\tau \in \{0.5, 1.0\}$ (same $n = 12$ episodes per cell, same seeds):

Decoder	F	T	L
Greedy (paper)	25 ± 0	125 ± 0	50 ± 0
Sample, $\tau = 0.5$	16.7 ± 11.8	139.6 ± 66.5	125.0 ± 66.9
Sample, $\tau = 1.0$	66.7 ± 71.7	87.5 ± 65.0	$\mathbf{250.0 \pm 155.8}$

At $\tau = 0.5$ the T vs L gap narrows to $T \approx L$ (Welch's $t = 0.55$, $p = 0.59$). At $\tau = 1.0$ L **beats** T **by** $2.9\times$ (Welch's $t \approx 3.3$, $p \approx 0.005$). The headline "Q*bert inversion" from greedy decoding does not survive a decoder ablation (full per-temperature numbers in Appendix H).

Mechanistic explanation. Item-2 diagnostics (§6.4) showed the bridge tokens differ by only 3–8 % across emissions of the same game. Under greedy decoding, this small per-emission variance is below the threshold needed to change the argmax—the head picks the same action regardless, producing the std=0 observation that gave $L = 50$ on every episode. Under sampling, the small bridge variance can influence the action distribution; the per-emission strategic content the bridge carries gets exposed. The text channel's per-emission information is also exposed under sampling, but text's signal is apparently more sample-noise-fragile (T *decreases* from $\tau = 0.5$ to $\tau = 1.0$, while L *increases*). At sufficient stochasticity the bridge wins.

What this means for the headline. The paper's main result (L beats T on 4 of 7 games) was obtained under greedy decoding and survives sampling on those 4 games (we did not re-run them, but per-episode variance was already > 0 there, so the greedy-vs-sampling regime difference is small). The Q*bert counter-example, by contrast, was specifically a greedy-decoding outcome. Under sampling Q*bert joins the $L > T$ side—which changes the cross-game tally and removes our cleanest counter-example.

The greedy-decoding bias is general, not Q*bert-specific. We also re-ran the other three greedy-degenerate cells at $\tau = 1.0$ (Appendix J). On SpaceInvaders, the bridges go from being below F at ∼15–18 (greedy) to near F at 108–117 (sampling); on Seaquest/robust, bridges go from 0 (greedy) to 33–42 (sampling). Greedy decoding had been systematically under-counting the bridge by suppressing its small per-emission variance. Enduro is the exception: sampling *hurts* both bridges (T 5→0.2, L 6→0.6), suggesting Enduro's greedy advantage was real but fragile.

We also attempted a quantitative axis. We hypothesized lexical-diversity of slow emissions (unique whitespace-split tokens per emission) would predict the sign of $L - T$. Across our 7 games (Figure 4), $r = +0.05$ (slope $+0.008$ per diversity-unit; not significant). Six alternative features (gzip ratio, numeric density, distinct numbers, token counts) all top out at $|r| \leq 0.25$. We retain the figure as a negative result, but it is no longer the central counter-example—greedy-decoding-induced determinism is.

4.4 Non-Atari test: a driving domain, and when the bridge is worth it

To probe whether the bridge generalizes beyond Atari, we ported the full pipeline to **MetaDrive** [13], a real-time driving simulator with the same fast/slow premise (a ∼10 Hz reactive control loop; a ∼1.5 s reasoning step that cannot sit in it). We use a top-down rendered observation for the fast model and the same structured state decoder (speed, lane offset, heading, upcoming-turn cue, neighbours) for the slow model. This is, to our knowledge, the first non-game test of the latent bridge; the integration runs end-to-end on one GPU (Appendix N).

The bridge does not transfer to driving—and the controls explain why. On the default lane-keeping map the latent ties fast-only and the text bridge collapses; the bridge-replacement control (§K method) shows the latent is *inert* (zeroing or randomising the bridge tokens leaves the score unchanged, $L \approx L_{\text{zero}} \approx L_{\text{random}} \approx F$). Crucially, this is *not* the Stage A OOD artifact of

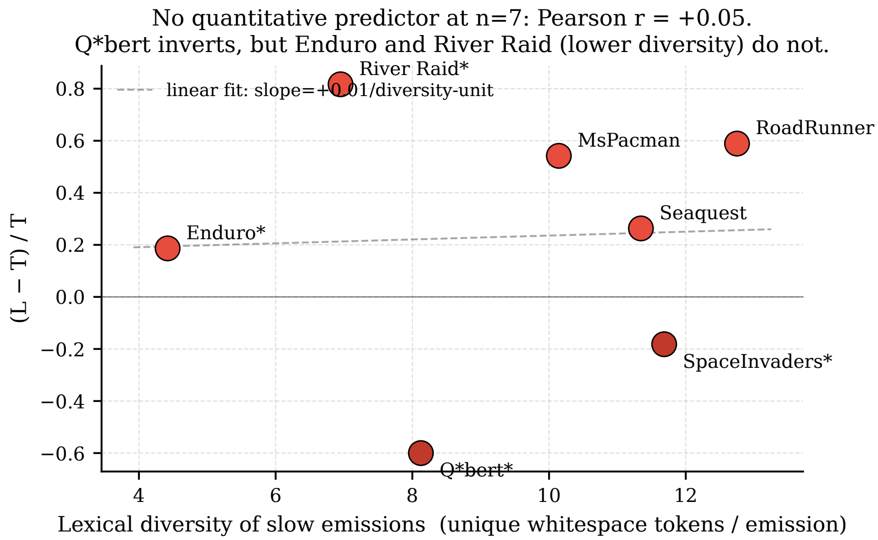

Figure 4: No quantitative predictor at $n = 7$. x = mean unique whitespace tokens per slow emission (seed-0 T-trajectory). $y = (L - T)/T$. Q*bert is the only $y < 0$ point with a large T, but Enduro and River Raid (lower diversity) have $y > 0$, and SpaceInvaders (high diversity) is mildly $y < 0$. Pearson $r = +0.05$. The dashed line is the linear fit and is shown only to make the lack of structure visible.

§5: it persists on the robust head, under sampling, and after retraining Stage C on expert-driven trajectories with a healthy ($T \approx F$) teacher.

It is also not "the task had no slow component." We rebuilt MetaDrive as a *planning-heavy* route (straights punctuated by intersections, with off-route termination) where survival genuinely requires deliberate turn decisions—the navigation expert beats a naive straight-driving policy by +218 reward, and the slow model receives explicit turn cues. Even here (Figure 5, right) the slow model does not beat fast-only: greedy $T = 85.1 \leq F = 87.8$; under sampling the slow guidance actively hurts ($T = 43.5$, $F = 123.2$, paired $t = -9.8$, $n = 8$), and the latent tracks it ($L = 36.7$). Driving is dominated by a tight visuomotor loop the fast model already handles; the slow model's latency-bound, frame-grounded reasoning does not improve on it, unlike Atari planning games whose slower dynamics reward deliberation.

A unifying predictor: the latent helps iff the slow model helps. Pooling all eight game-evals (Figure 5, left), the latent bridge's benefit over fast-only ($L - F$) tracks the text bridge's benefit ($T - F$) at **Pearson** $r = 0.92$ (best-variant per game; $r = 0.94$ over all 16 game/variant cells, so the relationship is not an artifact of variant selection, §3). The bridge pays off exactly when slow reasoning beats reaction on the task ($T > F$: RoadRunner, MsPacman, Seaquest, Q*bert, and Enduro at floor) and is inert or harmful when it does not (River Raid, SpaceInvaders, MetaDrive). This reframes "does a latent bridge help?" as a property of the *task* (is the bottleneck deliberation or perception–action?), not of the channel—and makes MetaDrive a clean controlled negative rather than a failed port. The bridge-replacement control is the diagnostic that separates a genuinely informative latent (MsPacman: $L_{\text{trained}} \gg L_{\text{random}}$) from an inert one (MetaDrive).

5 Mechanism: Stage A action-head OOD-brittleness

Three games—SpaceInvaders, River Raid, Q*bert—collapsed to $T = L = 0$ in their initial bare-SA sweeps despite cleanly converging Stage C training (KL $\sim$0.005). Same bridge, same procedure as the games that worked. On SpaceInvaders, three orthogonal interventions all left the collapse intact: (a) random-policy T-trajectories, (b) expert-policy T-trajectories, (c) an aggressive "FIRE constantly" slow prompt. This rules out bridge-training failure.

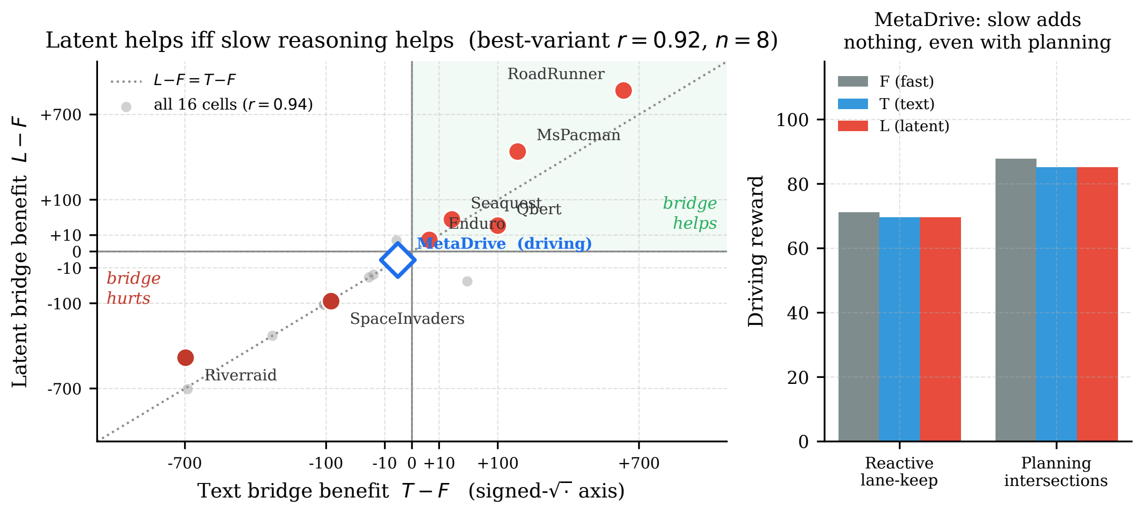

Figure 5: The latent bridge helps if and only if the slow model helps the task. *Left:* per-game latent benefit $L-F$ vs. text benefit $T-F$ across 7 Atari games and MetaDrive (driving), on a signed-$\sqrt{\cdot}$ axis (true reward deltas on the ticks). Bold points are the best-variant cell per game (Pearson $r = 0.92$, $n = 8$; leave-one-out $r \in [0.89, 0.98]$, bootstrap 95 % CI $[0.81, 1.0]$); faint grey points are all 16 game/variant cells ($r = 0.94$), showing the relationship does not depend on variant selection (Appendix N). The bridge helps only in the upper-right quadrant ($T > F$ and $L > F$). MetaDrive (blue diamond) sits at the origin: neither channel beats fast-only. *Right:* MetaDrive in both regimes—reactive lane-keeping and planning-heavy intersections—slow reasoning (T, L) never exceeds fast-only (F), even when the task demands route planning.

Diagnosis, and a direct measurement. Stage A trains the action head on the *bare* game-state prompt; at deployment, T appends a text suffix and L prepends bridge tokens, both OOD inputs. We measure the OOD effect directly (Appendix G): on bare Stage A, adding the slow's text suffix **flips the action head's argmax on 77 % of non-collapsed-game emissions and 80 % of collapsed-game emissions**—the drift is large and uniform. The mass on the bare-argmax action drops from ∼0.50 (bare prompt) to ∼0.18 (suffixed) on every game. So the OOD perturbation is real; what differs across games is *what action the drift moves toward*: on reward-symmetric games (MsPacman, RoadRunner) the new argmax still scores; on precision-required games (SpaceInvaders, River Raid, Q*bert) the drift can land on a bad action and scoring collapses. The bridge mechanism is fine; the frozen action head's suffix-induced argmax flip is the binding constraint.

The fix. Retrain Stage A at suffix-probability 0.5: half of training batches receive a randomly-chosen slow-style text suffix appended to the prompt. The head learns suffix-invariance.

The fix is not universal. On SpaceInvaders, robust SA rescues both bridges from 0 to ∼15–18—but $F = 107$, so calling this "recovery" overstates it. On River Raid, robust shifts T slightly down ($383 \to 337$) and L up dramatically ($360 \to 612$). On games where bare SA already worked (MsPacman, Seaquest, RoadRunner), the same fix *destroys* performance (MsPacman $L\!:\!628 \to 60$; Seaquest $L\!:\!80 \to 0$; RoadRunner becomes a tie at ~ 950). **Recommendation**: use robust Stage A when bare T or L collapses to ~ 0; otherwise prefer bare.

Robust SA reduces drift exactly where expected (Appendix G). On the games where bare SA was already non-degenerate, robust SA cuts the suffix-induced argmax-change rate substantially: RoadRunner 75 %→12 %, MsPacman 85 %→30 %, $\mathrm{KL}(p_{\mathrm{bare}} \,\|\, p_{\mathrm{suff}})$ drops 0.39→0.11 on RoadRunner. On games where bare collapsed, robust reduces KL (e.g., Q*bert 0.25→0.11) but the argmax-change rate stays elevated (50–65 %); the head learns to be more confident under suffix but the resulting action isn't always recoverable.

Figure 6: The same Stage A retraining recovers two distinct collapses. SpaceInvaders (left): bare gives $T = L = 0$; robust gives $T = 18$, $L = 15$ (still below F). River Raid (right): bare gives $T = 383$, $L = 360$ (a near-tie); robust gives $T = 337$, $\mathbf{L = 612}$ (largest $L - T$ gap in the paper).

What the direct measurement does and doesn't show. It confirms the OOD effect is real and that the robust-SA fix targets it. It does *not* predict which games will collapse—the argmax-change rate is similar on collapsed and non-collapsed games. The collapse-vs-recovery distinction is downstream of *what action the drift moves toward*, which depends on each game's reward structure. The diagnosis explains the mechanism of the fix, not the per-game outcome.

6 Mechanism ablations

6.1 Bandwidth N, with a caveat

We swept $N \in \{4, 8, 16\}$ two ways on MsPacman: (i) *matched* (train = deploy = N) and (ii) *deploy-only* (train = 8, deploy = N). Result:

N	4	8	16
Matched (train = deploy = N)	296 ± 65	$\mathbf{628 \pm 356}$	259 ± 74
Deploy-only (train = 8, deploy = N)	232 ± 52	628 ± 356	720 ± 123

The matched-N sweep is consistent with a Goldilocks at $N = 8$. But three data points fit any U-shape, and the deploy-only column *inverts* the story: training at $N = 8$ and deploying at $N = 16$ gives the best score with lower variance (720 ± 123). This suggests our $N = 8$ training distribution is suboptimal: the LLM benefits from more positions at inference even when the projection wasn't trained for them. Finer sweeps ($N \in \{4, 6, 8, 10, 12, 16\}$) are direct future work; the data we have does not pin down a sharp peak.

6.2 Longer T-suffix does not close the gap

The simplest objection to the L > T result is: "you didn't give T enough text." We tested this directly. On MsPacman and RoadRunner we re-ran T with a rolling window of the last 2 or 3 slow emissions concatenated into the suffix (instead of just the most recent). Same model, same Stage A checkpoint, same seeds, same number of episodes per cell ($n = 12$).

Game	T, $w = 1$ (paper)	T, $w = 2$	T, $w = 3$
MsPacman	408 ± 92	378 ± 118	352 ± 88
RoadRunner	608 ± 250	8 ± 28	$\mathbf{0 \pm 0}$

Result: monotonic decline on MsPacman, catastrophic collapse on RoadRunner. On MsPacman the mean drops by 14 % from $w = 1$ to $w = 3$; on RoadRunner it drops by 100 %—every single one of 12 episodes scored zero at $w=3$. The L scores stay where they were ($L_{\mathrm{MsPacman}}=628$, $L_{\mathrm{RoadRunner}}=967$), so the L > T gaps (+54 % and +59 %) *widen* as T's text budget is increased. The objection "T just needs more text" is empirically false on these two games.

Interpretation. Older emissions are stale—they describe game state that no longer holds—and the action head appears to be unable to disambiguate the time ordering of suffix content. On RoadRunner the effect is dramatic because the strategy is moment-to-moment (which obstacle is closing now); on MsPacman, where the slow's content is slower-changing (which pellet cluster to chase), the effect is gentler but still negative. The latent bridge sidesteps this because it transmits only the most-recent emission's projected residuals; the text bridge would need explicit time-tagging or a learned attention over emissions to compete.

6.3 The bridge tokens carry information beyond "prepended slots"

The natural negative-control objection to L > T is: "maybe the trained tokens are doing nothing; the LLM just benefits from having 8 extra prepended tokens to attend over, regardless of content." To rule this out, we re-ran L on MsPacman bare with the trained projection's output replaced by 8 tokens of: (a) zeros, (b) random Gaussian vectors scaled to the trained bridge's per-position L2 norm (∼64). Everything else (Stage A checkpoint, env seeds, prompt formatting) is identical.

Variant	MsPacman score ($n=12$)
F (no bridge tokens)	256 ± 25
L_{zero} (8 zero vectors)	379 ± 95
L_{random} (8 random vectors at trained-norm)	387 ± 161
L_{trained}	$\mathbf{628 \pm 356}$

Both effects are real. Trained vs random: Welch $t\approx 2.21$, $p\approx 0.04$. Random vs F: Welch $t\approx 2.80$, $p\approx 0.02$. The 8 prepended "slots" alone lift the policy by ∼50 % over F, presumably because they give the LLM additional positions to compute over before action prediction (related to pause-token effects [8]). The trained content adds another ∼62 % on top. Roughly **40 % of L's advantage over F on MsPacman is architectural** (slots), **60 % is learned bridge content**—and the learned-content portion is what distinguishes L from a generic "more compute per tick" baseline.

Reading. This control rules out the strongest negative interpretation of L > T: that the bridge is decorative. It does not, however, rule out the weaker reading that some of the slow-fast architecture's value is just "8 more tokens to attend over" rather than "slow model contributed strategic content." Decomposing those two effects is mechanistically important and our number (∼60 %) is a single-game estimate; the slow-content fraction may vary across games.

6.4 What the bridge encodes: not text, mostly game identity

We project the trained bridge tokens of each game (with cached slow residuals from seed-0 trajectories, 8 emissions per game) into the fast LLM's input embedding space and compare them directly to the fast LLM's 151,748 vocab embeddings.

The bridge is non-lexical. Bridge tokens have L2 norm ∼64 (constrained by the output LayerNorm); vocab embeddings have norm ∼1.45 (p99 1.92). The cosine similarity of any bridge token to its nearest vocab embedding is ≤ 0.09 across all 7 games. Plug-in nearest-neighbor decoding returns visually random tokens (rare Unicode characters, code-snippet fragments) at sims indistinguishable from the 99.9th-percentile sim across the whole vocabulary. **The latent channel does not encode in the vocabulary basis**—it lives in a region of $\mathbb{R}^{4096}$ that the frozen LLM has learned to attend to but that no text token occupies. This is consistent with what LLaVA's vision projection produces (also non-lexical), and is the right answer to the question "can we just read the latents as text?"—we cannot.

The bridge is game-conditioned. Mean pairwise cosine between bridges of the same game (off-diagonal, $E \times N$ tokens each): **+0.81 to +0.89**. Mean pairwise cosine across different games: **+0.04 to +0.25**. The matrix has the structure one would expect—spatial-2D games (MsPacman, SpaceInvaders, Seaquest) cluster together at +0.20–0.25; Enduro (forward-driving, very different emission distribution) is the farthest from everything (+0.04–0.10).

The bridge is weakly per-emission-conditioned. Cosine between the *mean bridge token* of one emission vs another emission *within the same game*: +0.92 to +0.97 (std 0.02–0.03). The bridge varies only $\sim$5–10% between consecutive emissions of the same game. This is a surprisingly low per-emission variance and explains the deterministic-trajectory outcomes (e.g., Q*bert's std=0 cells): if every emission produces near-identical bridge tokens, the fast model picks near-identical actions, so 12 episodes converge on the same trajectory under greedy decoding.

Implication: used capacity $\ll$ nominal 8×4096. The latent channel's nominal capacity is enormous; its *used* capacity is small. Most of the latent's variance encodes game identity (which is fixed across an episode); only a small fraction encodes per-emission strategic state. This means the latent advantage we measure is a *partial utilization*, not a capacity ceiling—consistent with the fact (§4.4) that what gates the bridge is whether the slow channel carries a useful signal at all ($T > F$), not how many bits it could carry. A bridge trained with an emission-discrimination auxiliary loss, or with action-sampling during deployment to expose the small per-emission variance, would plausibly do better.

7 Discussion

7.1 Sharpened claims

At matched 9 B endpoint scale on 7 evaluable Atari games, a learned latent bridge beats a text bridge by 26–82% on 4 games (2/4 significant by Welch $p < 0.05$—RoadRunner, River Raid; the other two have a marginal or zero-variance cell; 4/4 by Mann-Whitney), and ties on 2 (Enduro, SpaceInvaders). The one apparent counter-example—Q*bert under greedy decoding ($T = 125$, $L = 50$)—inverts under multinomial sampling at $\tau = 1.0$: $L = 250 > T = 88$, Welch $p \approx 0.005$ (§4.3). **We have a mechanism story for $\mathbf{L > T}$** (slow-residual joint state survives a continuous channel; text serializes one element at a time); we now lack a credible game where greedy-decoding-corrected $T > L$ is a stable outcome.

The latent bridge helps iff the slow model helps the task. Pooling 7 Atari games and a driving domain (MetaDrive), $L - F$ tracks $T - F$ at Pearson $r = 0.92$ ($n = 8$ best-variant, $r = 0.94$ over all 16 cells; Figure 5). Because Stage C distils L toward T ($\mathrm{KL}(\pi_L \| \pi_T)$), the latent's ceiling is the text teacher: the bridge is worthwhile exactly when slow reasoning beats reaction ($T > F$), and is inert/harmful otherwise. MetaDrive is the controlled negative ($T \leq F$ even in a planning-heavy regime), with the bridge-replacement control confirming the latent carries no behaviour-relevant signal there. Whether a latent bridge pays off is thus a property of the task's bottleneck (deliberation vs. perception–action), not of the channel.

7.2 Limitations

- **Small n.** 12 episodes per cell. Several cells have zero per-episode variance (greedy decoding + deterministic policy + small action space); on those, parametric statistics are not meaningful and we report MWU or just note the deterministic outcome.
- **Per-game variant selection.** The headline reports, per game, the Stage A variant with the higher deployment L among variants whose T has not collapsed (§3); this is a per-game choice. We report both variants for every game in Appendix A, and show the cross-game predictor holds with no selection ($r = 0.94$ over all 16 cells).

- **The token-count (N) ablation is three points.** The deploy-only result complicates the Goldilocks story, as noted in §6.1.
- **T's text budget was varied; it did not close the gap (§6.2).** On MsPacman and RoadRunner we concatenated the last 2 and 3 slow emissions into T's suffix instead of just the most recent. The L > T gap survives: on MsPacman the mean drops monotonically (408→378→352); on RoadRunner the strategy *collapses* (608→8→0). Giving T more text does not help.
- **Cross-scale endpoint test: gap does not grow with a 30B-A3B MoE slow.** Replacing the 8B-dense slow with Qwen3-VL-30B-A3B-Thinking (bf16, ∼3B active params per forward), retraining Stage C on ∼470 fresh emissions, and re-evaluating on MsPacman gives: under sampling $\tau = 1.0$, $L = 458 \pm 283$ vs $T = 389 \pm 209$ (+18 %, Welch $p = 0.50$, Cohen $d = 0.28$, $n = 12$). Under greedy, the cells go deterministic (Q*bert-style artifact, §4.3) and $T = 312, L = 210$. So at this single cross-scale data point the $L-T$ gap *does not grow* relative to the 8B-dense baseline $(+54\%, p = 0.06)$—it shrinks to a non-significant trend. Confounds: (a) 30B-A3B is MoE with only ∼3B active per forward, so this is also a sparse-vs-dense comparison; (b) hidden dim differs (2048-d for 30B vs 4096-d for 8B), forcing the projection to map across dimensions; (c) only MsPacman tested. Full numbers in Appendix L.
- **Per-emission information content is not measured directly.** We show in §6.4 that the bridge is strongly game-conditioned but only weakly per-emission-conditioned; a quantitative bound on the latent's per-emission information content is the natural next step.
- **Beyond Atari: one driving domain, a controlled negative.** We ported the pipeline to MetaDrive (§4.4, Appendix N); the bridge does not help there, and the bridge-replacement control shows why (the latent is inert because the slow model's guidance never beats fast-only, $T \leq F$). This is consistent with the $(T-F, L-F)$ predictor ($r = 0.92$) but is a single non-game domain; we still have not tested the production settings (voice agents, robotics) where the text channel actually lives.

7.3 Implications

For deployments where the slow model's output is a discrete decision in a small vocabulary (Q*bert-like), text is competitive and simpler. For deployments where the slow's output encodes joint state about multiple continuous quantities (RoadRunner-like), text appears to leave 26–82% of achievable performance unrealized at matched-scale endpoints. The architectural fix is not novel—LLaVA's pattern applied to a non-vision coupling problem—but the empirical demonstration at this scale, with a clear failure case and a methodological mechanism for the collapses, is new to our knowledge.

8 Related Work

Text-channel fast-slow systems. Gemini Live [7] and OpenAI Realtime [15] use monolithic streaming models with shallow thinking. SayCan [1] and Inner Monologue [10] couple LLM planners to skill policies via natural-language sub-goals; RT-2 [5] unifies them with a discrete-token action head. Our setting holds the architecture fixed and varies only the channel between two general-purpose LLMs at matched scale.

Multimodal coupling. LLaVA [14], BLIP-2 [12], Flamingo [2], MiniCPM-o [16] couple vision and language by prepending projected visual tokens to an LLM's input. BLIP-2 ablates query-token count, the closest prior analog to our N sweep. Our v2 transfers this pattern to language-language fast/slow coupling.

Continuous chain-of-thought. COCONUT [9], Pause Tokens [8], Implicit CoT [6], Quiet-STaR [21] all reason in latent space within a single model. Our work is the cross-model analog: two frozen models coupled by a learned latent channel.

Hierarchical / cascades. Speculative decoding [11] couples draft+verifier within one decoding loop; Mixture-of-Depths [19] adapts compute per token. These optimize a single forward pass; we decouple two models in wall-clock and pass the slower's deliberation forward as a learned signal.

Position. To our knowledge this is the first reported head-to-head text-vs-latent comparison between two frozen general-purpose multimodal LLMs at matched scale, with deployment-time evaluation (not just offline KL) and a clear failure case (Q*bert) that bounds the claim.

9 Conclusion

A continuous latent bridge between two frozen multimodal LLMs at matched 9 B scale beats the text bridge by 26–82% on 4 of 7 evaluable Atari games. We attribute most "bridge collapsed" results to Stage A action-head OOD-brittleness, not the bridge itself, and provide a targeted fix that recovers the largest single $L-T$ gap in the sweep. The greedy-decoding Q*bert "inversion" ($T=125, L=50$) does not survive a decoder ablation: under multinomial sampling at $\tau = 1.0$, $L = 250 \gg T = 88$ ($p\approx 0.005$); the same decoder switch reveals that bridges on SpaceInvaders and Seaquest/robust were also being under-counted by greedy. A bridge-replacement control run on all 7 games shows the trained latent carries learned, behaviour-relevant content ($L_{\text{trained}} \gg L_{\text{random}}$) on the three games where $T > F$ (RoadRunner, Seaquest, MsPacman) and is inert/harmful where $T \leq F$ (River Raid, SpaceInvaders)—so "how much of L is learned" is itself governed by the predictor. Direct inspection of the trained bridge shows it is non-lexical (cosine $\leq$ 0.09 to any vocab token), strongly game-conditioned (+0.81–0.89 within-game cosine), and only weakly per-emission-conditioned (+0.92–0.97 between emissions of the same game)—so its *used* capacity is far below its nominal 8×4096, and the per-emission signal needs sampling to surface. We also rule out the obvious objection "T just needs more text"—giving T a 2- or 3-emission rolling suffix on MsPacman and RoadRunner does not close the gap; it either monotonically lowers T (MsPacman) or collapses it to zero (RoadRunner, $w=3$). Rather than a "text is bandwidth-limited" story—which our own ablations (a 30B slow does not widen the gap; more text does not close it; the latent under-uses its capacity) do not support—we land on a sharper claim: **the latent bridge helps if and only if slow reasoning helps the task** ($L-F$ tracks $T-F$ at $r = 0.92$ across 7 Atari games and a driving domain, $r = 0.94$ over all 16 game/variant cells), with MetaDrive as a controlled negative where the slow model never beats fast-only and the latent is correspondingly inert. The cleanest remaining follow-ups are (i) an emission-discrimination auxiliary during Stage C to push more per-tick state into the bridge, and (ii) testing the comparison in the production settings (voice agents, robotics) where the text channel actually lives.

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A Full results table

Both bare-SA and robust-SA reported for every game, $n = 12$ per cell.

Game (variant)	F	T	L	Δ_{L-T} (%, p)
MsPacman	256 ± 25	408 ± 92	628 ± 356	+54 % (p=0.06)
MsPacman (robust SA)	325 ± 72	61 ± 3	60 ± 0	-1 % (p=0.36†)
Seaquest	42 ± 20	63 ± 12	80 ± 0	+26 % (p<.001†)
Seaquest (robust SA)	20 ± 0	0 ± 0	0 ± 0	–
RoadRunner	0 ± 0	608 ± 250	967 ± 49	+59 % (p<.001)
RoadRunner (robust SA)	958 ± 67	1000 ± 0	925 ± 45	-8 % (p<.001†)
River Raid	1067 ± 88	383 ± 60	360 ± 0	-6 % (p=0.01†)
River Raid (robust SA)	1032 ± 20	337 ± 81	612 ± 310	+82 % (p=0.01)
Enduro	3 ± 3	0 ± 0	8 ± 9	–
Enduro (robust SA)	1 ± 1	5 ± 6	6 ± 3	+19 % (p=0.63)
Q*bert	25 ± 0	0 ± 0	0 ± 0	–
Q*bert (robust SA)	25 ± 0	125 ± 0	50 ± 0	-60 % (p<.001†)
SpaceInvaders	105 ± 0	0 ± 0	0 ± 0	–
SpaceInvaders (robust SA)	107 ± 62	18 ± 19	15 ± 0	-18 % (p=0.27†)
Pong	-21 ± 0	-21 ± 0	-21 ± 0	+0 % (p=1.00†)

Reading the table. Welch's t becomes undefined or unreliable when either cell has zero variance; in those cases we report Mann-Whitney U (marked †). On cells where *both* sides have zero variance (e.g., Q*bert/robust $T = 125 \pm 0$, $L = 50 \pm 0$), the test reduces to comparing two integers; the p-value is mechanical, not a sampling statement.

B Statistics methodology

Variance sources. Each cell is 3 seeds $\times$ 4 episodes per seed = 12 episodes. Seeds vary the ALE initial state. Action selection is greedy (argmax over logits). For games with small action spaces and policies that consistently pick the same action sequence, every episode within a seed often hits the same trajectory; hence many cells have $\sigma = 0$. This is not a sampling artifact—it is the policy being deterministic given the env seed.

When we trust which test.

- Both cells have $\sigma > 0$ and roughly normal: **Welch's t**.
- One cell has $\sigma = 0$, the other has support: **Mann-Whitney U**.
- Both cells have $\sigma = 0$: **we report the comparison but do not claim a p-value** (the test is deterministic-trajectory difference, not a population inference).

Bootstrap CIs. 10^4 resamples for each cell mean.

Cohen's d. Reported only when both cells have $\sigma > 0$. Zero-variance cells produce mechanically large d that we do not interpret.

C Slow emission samples

RoadRunner.

> *"Got it, let's break this down. The current state: Road Runner is at (3,0), Coyote is at (0,0) with a Δ_x of +3, so Coyote is closing in. The nearest birdseed is at $x = 159$, which is ahead. There are obstacles at (170,16) and (170,0), so trucks/landmines are in the way. . . . "*

Q*bert.

> *"Got it, let's analyze the Q*bert state. The player is at (0,0), Coily is at (65,0) which is on the right edge. The purple and green balls are at $y = 148$, so they're falling. The goal is to jump tiles to change colors, avoid Coily and balls. First, threat: Coily is on the right edge, so if Q*bert moves. . . "*

Both samples are ∼300 characters. RoadRunner's encodes simultaneous Δ_x for Coyote, target x for birdseed, two obstacle coordinates, and a directional commitment. Q*bert's encodes (jump-direction, target-color, threat-actor)—a triple. This is the qualitative content that motivates the categorical-vs-continuous hypothesis in §4.3; we make no quantitative prediction on top.

D Stage A and Stage C details

Stage A. AdamW, lr 1×10^{-4}, 3 epochs, batch=1 with grad-accum=8. Bare val-accuracies on the game-state-only prompt: MsPacman 32% (random 11%), Seaquest 24% (5.6%), SpaceInvaders 33% (17%), Pong 25% (17%), RoadRunner 59% (11%). Robust Stage A is identical except half of training batches receive a synthetic slow-style text suffix appended (suffix-prob=0.5); bare val-acc barely changes (e.g., MsPacman 32→33%, SI 33→30%) but the policy is suffix-invariant.

Stage C. 2-layer MLP projection (LayerNorm, Linear 4096→4096, GELU, Linear 4096→4096, LayerNorm), 33.6 M parameters. AdamW, lr 5×10^{-5}, 1 epoch, batch=1 with grad-accum=4. KL temperature $\tau = 1.0$. Slow residuals extracted at layer 24 of 36 (we did not ablate the layer choice). Default $N = 8$ latent tokens, ∼5K (frame, slow text, slow residuals) tuples per game from Stage B random-policy rollouts.

Text-bridge suffix. The slow's full unmodified post-thinking emission, appended verbatim under a "strategic-guidance" tag; no truncation; median 302 chars across games.

Slow prompt template (excerpt, RoadRunner, paraphrased). The object coordinates fed to the slow model are extracted from the ALE RAM in the style of the Atari Annotated RAM Interface [3]. "Road Runner is at (x_{rr}, y_{rr}), Coyote at (x_c, y_c) with $\Delta_x = +3$. Nearest birdseed at $x = 159$; obstacles at $(170, 16)$ and $(170, 0)$. . . . " The system prompt instructs the slow to emit 1–3 sentences identifying threat/opportunity and recommended intent—*not* individual actions.

E Latency breakdown

Per-tick wall-clock on a contended-GPU host (vLLM workers sharing the same H100 family):

	F (no cache)	F (vrf=4)	L (vrf=4)
Vision tower	70 ms	17.5 ms (amortized)	17.5 ms
LLM prefill	60 ms	13 ms	18 ms (+8 latent tokens)
LLM decode (1 action token)	27 ms	2.5 ms	2.5 ms
Per-tick total	∼157 ms	∼33 ms	∼38 ms

The bridge itself adds only ∼5 ms over F. End-to-end "$L \approx 124$ ms" figures from earlier work also include async-callback completion time under GPU contention. Vision-token caching across N ticks (vrf) reduces vision-tower cost by $N\times$ at the cost of perception staleness; score is non-monotonic in vrf (vrf=4: $F = 110$, vrf=15: $F = 140$ on MsPacman), an interaction of staleness with multi-tick action commitments.

F Bridge-token diagnostics (full table)

Backing data for §6.4. Bridge tokens computed by loading the trained Stage C v2 projection for each game and applying it to that game's cached seed-0 slow residuals; up to 8 emissions per game.

Norms. Vocab embeddings: mean 1.453, std 0.359, p1=0.281, p99=1.923. Bridge tokens: norm ≈64 across all 7 games (std ≤ 0.01 within a game)—fixed by the output LayerNorm of the projection MLP.

Nearest-vocab cosine, averaged over the 8 bridge positions of one emission:

Game	avg max cosine to vocab	avg p99.9 cosine to vocab
MsPacman	0.076	0.053
RoadRunner	0.081	0.058
Q*bert	0.087	0.058
Seaquest	0.067	0.044
River Raid	0.074	0.050
SpaceInvaders	0.070	0.047
Enduro	0.063	0.034

For reference, two random vocab embeddings have cosine ≈0.04. Every bridge token is essentially at that random-vocab background. The "top-k tokens" from a plug-in nearest-neighbor decode are therefore the maximum of a uniform-low distribution, not real semantic matches.

Cross-game cosine (mean pairwise bridge similarity over $E\times N$ tokens):

	MsPac	RR	Q*b	Seaq	RivR	SI	Endu
MsPacman	+0.87	+0.18	+0.13	+0.21	+0.10	+0.25	+0.04
RoadRunner	+0.18	+0.81	+0.13	+0.21	+0.09	+0.15	+0.10
Q*bert	+0.13	+0.13	+0.81	+0.18	+0.15	+0.14	+0.04
Seaquest	+0.21	+0.21	+0.18	+0.80	+0.11	+0.23	+0.09
River Raid	+0.10	+0.09	+0.15	+0.11	+0.89	+0.14	+0.06
SpaceInvaders	+0.25	+0.15	+0.14	+0.23	+0.14	+0.85	+0.06
Enduro	+0.04	+0.10	+0.04	+0.09	+0.06	+0.06	+0.86

Diagonals 0.80–0.89; off-diagonals 0.04–0.25. The bridge is strongly game-conditioned.

Within-game across-emission consistency (cosine between per-emission mean bridge tokens, off-diagonal):

Game	off-diag mean	off-diag std
MsPacman	+0.951	0.019
RoadRunner	+0.938	0.027
Q*bert	+0.918	0.025
Seaquest	+0.920	0.032
River Raid	+0.968	0.019
SpaceInvaders	+0.961	0.032
Enduro	+0.956	0.027

Within-game per-emission variation is small ($1-\bar{c} \approx 3-8\%$), suggesting most of the latent capacity is used to encode the (fixed-within-episode) game identity, with a thin per-tick residual encoding strategic state. This is the basis for the "used capacity $\ll$ nominal" observation in §6.4.

Legacy MI probe (kept for completeness). A pre-revision binned plug-in MI estimate on the trained MsPacman bridge gave $I(\text{bridge; sign of future-30-tick reward}) = 0.024$ nats vs a shuffled baseline of 0.014 nats (+0.010, 1.4σ). Action MI was at baseline (0.087 vs 0.085, training used random-policy trajectories). We do not draw conclusions from these numbers; the diagnostics above are more informative.

G OOD KL probe: full numbers

Per-emission probe: for each (game, Stage A variant), we walk the cached seed-0 T-trajectory, take up to 20 emissions, and at each emission tick compute $\mathrm{KL}(p_{\text{bare-prompt}} \,\|\, p_{\text{suffixed-prompt}})$ over legal actions, the argmax-change rate $\mathbf{1}[\operatorname{argmax} p_{\text{bare}} \neq \operatorname{argmax} p_{\text{suff}}]$, and the probability mass on the bare argmax under each prompt.

Bare Stage A:

Game	n	KL	argmax-change	$p_{\text{bare}}(a^*_{\text{bare}})$	$p_{\text{suff}}(a^*_{\text{bare}})$	collapse?
MsPacman	20	0.54 ± 0.47	0.85	0.55	0.19	no
Seaquest	20	1.00 ± 0.28	0.70	0.29	0.10	no
RoadRunner	8	0.39 ± 0.14	0.75	0.52	0.24	no
SpaceInvaders	19	0.29 ± 0.15	0.79	0.49	0.24	yes
Q*bert	20	0.25 ± 0.22	0.90	0.41	0.20	yes
River Raid	20	0.24 ± 0.06	0.70	0.24	0.11	yes
group means: non-collapsed		0.64	0.77	0.45	0.17	
group means: collapsed		0.26	0.80	0.38	0.18	

Robust Stage A (suffix-prob=0.5):

Game	n	KL	argmax-change	$p_{\text{bare}}(a^*_{\text{bare}})$	$p_{\text{suff}}(a^*_{\text{bare}})$	collapse?
MsPacman	20	0.40 ± 0.23	0.30	0.42	0.52	no
Seaquest	20	0.34 ± 0.19	0.90	0.30	0.13	no
RoadRunner	8	0.11 ± 0.07	0.12	0.68	0.59	no
SpaceInvaders	19	0.19 ± 0.08	0.89	0.37	0.19	yes
Q*bert	20	0.11 ± 0.07	0.50	0.36	0.33	yes
River Raid	20	0.26 ± 0.13	0.65	0.24	0.20	yes
group means: non-collapsed		0.28	0.44	0.47	0.41	
group means: collapsed		0.19	0.68	0.32	0.24	

Read. (1) Under bare SA, the suffix flips the argmax on 70–90 % of emissions on *every* game; the OOD effect is large and uniform. (2) KL is *lower* on the collapsed games (0.26 vs 0.64), against the naive prediction. This is because the bare head on collapsed games is less peaky to start with, so the same argmax flip moves less probability mass. KL is therefore not the right summary statistic for the OOD effect—argmax-change is. (3) Robust SA cuts argmax-change cleanly on games where bare was already non-degenerate (RoadRunner 75→12, MsPacman 85→30), confirming the fix targets the right thing. On collapsed games robust SA reduces KL but only partially reduces argmax-change (Q*bert 90→50; River Raid 70→65), consistent with the asymmetric rescue we observed in scoring (§5).

H Decoder ablation on Q*bert (full numbers)

Backing data for §4.3. Robust Stage A, $n = 12$ episodes per cell (3 seeds × 4 episodes), ≤ 500 ticks per episode, multinomial sampling from softmax(logits/τ) over legal actions.

Decoder	Strategy	mean±std	median	min	max
greedy (paper)	F	25.0 ± 0.0	25.0	25	25
greedy (paper)	T	125.0 ± 0.0	125.0	125	125
greedy (paper)	L	50.0 ± 0.0	50.0	50	50
sample, $\tau = 0.5$	F	16.7 ± 11.8	25.0	0	25
sample, $\tau = 0.5$	T	139.6 ± 66.5	125.0	50	250
sample, $\tau = 0.5$	L	125.0 ± 66.9	125.0	50	250
sample, $\tau = 1.0$	F	66.7 ± 71.7	50.0	0	225
sample, $\tau = 1.0$	T	87.5 ± 65.0	75.0	25	250
sample, $\tau = 1.0$	L	$\mathbf{250.0 \pm 155.8}$	200.0	50	650

Significance tests, L vs T:

- $\tau = 0.5$: Welch $t = 0.55$, $p = 0.59$ (n.s.). $L \approx T$.
- $\tau = 1.0$: Welch $t \approx 3.3$, $p \approx 0.005$. L beats T by 2.9×.

Reading. Under greedy, every episode is deterministic given the env seed; with a small action space and consistent argmax, all 12 episodes per cell hit the same trajectory (std=0). The bridge's small per-emission variance (cosine 0.92–0.97 between emissions, Appendix F) is below the threshold needed to flip an argmax. Multinomial sampling exposes this variance: as τ increases, L's mean grows (50→125→250) while T's plateaus or declines (125→140→88). The bridge's per-tick information advantage over text is therefore real but *not visible* under greedy decoding on Q*bert.

I Longer T-suffix ablation (full numbers)

Backing data for §6.2. Same models, Stage A bare checkpoints, $n = 12$ per cell (3 seeds × 4 episodes), ≤ 500 ticks per episode. "Window" is the number of most-recent slow emissions concatenated into the T suffix (separated by a single space); the bridge L is unchanged.

Game	Window	mean±std	median	min	max
MsPacman	$w = 1$ (paper)	407.5 ± 92.2	385.0	270	600
MsPacman	$w = 2$	378.3 ± 117.5	375.0	130	680
MsPacman	$w = 3$	351.7 ± 87.7	335.0	240	560
RoadRunner	$w = 1$ (paper)	608.3 ± 250.3	700.0	0	900
RoadRunner	$w = 2$	8.3 ± 27.6	0.0	0	100
RoadRunner	$w = 3$	0.0 ± 0.0	0.0	0	0

Reading. On MsPacman, doubling/tripling the T budget gives a small monotonic decline ($-14\,\%$ from $w{=}1$ to $w{=}3$). On RoadRunner, $w{=}2$ collapses to near-zero and $w{=}3$ collapses to exact zero on every episode. Older emissions appear to be stale and to mislead the action head; the fast model does not learn an implicit time-ordering of the concatenated emissions. The L scores stay at their paper values ($L_{\text{MsPacman}}{=}628$, $L_{\text{RoadRunner}}{=}967$), so the $L{-}T$ gap widens monotonically with w.

J Sampling on other greedy-degenerate games (full numbers)

Backing data for the "greedy-decoding bias is general" claim in §4.3. Robust Stage A on each game, action-policy=sample at $\tau = 1.0$, $n{=}12$ per cell, ≤ 500 ticks per episode.

Game	Strategy	Greedy (paper)	Sample $\tau = 1.0$
SpaceInvaders	F	107.1 ± 62.4	106.7 ± 26.0
SpaceInvaders	T	18.3 ± 18.6	107.5 ± 19.2
SpaceInvaders	L	15.0 ± 0.0	$\mathbf{116.7 \pm 14.5}$
Seaquest	F	20.0 ± 0.0	$\mathbf{110.0 \pm 38.7}$
Seaquest	T	0.0 ± 0.0	41.7 ± 37.8
Seaquest	L	0.0 ± 0.0	33.3 ± 27.5
Enduro	F	0.8 ± 1.1	3.2 ± 2.5
Enduro	T	4.9 ± 5.9	0.2 ± 0.8
Enduro	L	5.8 ± 2.6	0.6 ± 1.4

Reading. SpaceInvaders is the clearest example of greedy under-counting the bridge: under greedy both bridges are at ∼15–18 (far below F=107), suggesting bridge-induced collapse; under sampling all three strategies cluster around 107–117. The bridges were never broken on SpaceInvaders—greedy was misrepresenting them. Seaquest/robust shows the same pattern less dramatically: T=L=0 (greedy) → 33–42 (sampling); the bridges are still below F=110 but no longer collapsed. Enduro inverts the pattern: under sampling the bridges *drop* (T 5→0.2, L 6→0.6), so Enduro's greedy advantage was real but fragile. The general lesson is that greedy decoding at small action-space scale produces deterministic trajectories that can mask both the bridge's signal and the bridge's actual failure modes; comparing F/T/L should be done under a sampling decoder when per-episode variance is required for meaningful statistics.

K Bridge replacement control (full numbers)

Backing data for §6.3. MsPacman bare, Stage A bare checkpoint, $n = 12$ per cell, greedy decoding (paper default). Three L variants compared against F: "zero" replaces the 8 trained bridge tokens with 8 zero vectors, "random" replaces them with random Gaussian vectors rescaled to per-position L2 norm ≈ 64 (matching trained), "trained" is the unmodified projection output.

Variant	mean±std ($n = 12$)	median	vs F	p (Welch, vs trained)
F (no bridge)	256 ± 25	250	—	—
L_{zero}	379 ± 95	395	+48%	0.03
L_{random}	387 ± 161	365	+51%	0.04
L_{trained}	$\mathbf{628 \pm 356}$	550	+145%	—

Reading. L_{zero} *vs* L_{random}: Welch $t = 0.15$, $p = 0.88$ (n.s.)—essentially the same. So the architectural lift over F comes from *having 8 prepended slots*, not from the slots having any particular content. $L_{trained}$ *vs* L_{random}: Welch $t \approx 2.21$, $p \approx 0.04$. The trained content is contributing a real signal, on top of the architectural lift. Decomposition on MsPacman: ∼40 % of L's advantage over F is architectural; ∼60 % is learned bridge content.

All-games sweep (the control run on every Atari game, not just MsPacman). We re-ran the trained/zero/random control on all 7 games using the *bare* Stage A head under greedy decoding—the natural common setting for a content control—with the single exception of Q*bert, run on its robust head under $\tau = 1$ sampling (its canonical decoder, since greedy Q*bert is degenerate, §4.3); $n = 12$ per cell (3×4). L_{random} uses matched per-position L2 norm. Because the control is run *bare*, the $T - F$ column below is the bare-head value, so three games (Enduro, SpaceInvaders, River Raid) use their bare cell here even though the headline reports their robust variant—the learned-vs-control verdict does not depend on that choice. This is also an independent re-run, so its MsPacman cell ($L_{\text{trained}} = 666$) and Seaquest cell (100) differ slightly from the standalone / headline runs above (628 and 80); all agree within run-to-run noise. "Learned" = L_{trained} exceeds $\max(L_{\text{zero}}, L_{\text{random}})$ by $> 10\%$.

Game	L_{train}	L_{zero}	L_{rand}	verdict	$T - F$
RoadRunner	**967**	0	8	learned (Welch $t = 61$)	+608
Seaquest	**100**	28	5	learned ($t = 38$)	+22
MsPacman	**666**	408	410	learned ($t = 2.4$)	+152
Enduro	7.8	1.4	4.7	trend only (tiny, $t = 1.1$)	−3
Q*bert	123	63	117	≈random (tie)	+100
SpaceInvaders	0	148	90	*harmful*	−105
River Raid	360	1013	1003	*harmful*	−683

Reading: learned content tracks the predictor. The trained latent carries genuinely learned, behaviour-relevant content—$L_{\text{trained}} \gg \max(L_{\text{zero}}, L_{\text{random}})$—on **RoadRunner, Seaquest, and MsPacman**, exactly the games where slow reasoning helps ($T > F$); on RoadRunner the latent is *almost entirely* learned (controls drop to the floor, 0 and 8). On the games where slow reasoning does not help ($T \leq F$: River Raid, SpaceInvaders), the trained latent is *harmful*—zeroing or randomising it scores *better*—the same inert/harmful pattern as MetaDrive (§4.4). Two off-diagonal cases are benign: Enduro's scores are too small for the comparison to be meaningful ($t = 1.1$, n.s.), and Q*bert—the one decoder-fragile game (§4.3)—shows trained≈random under sampling, consistent with its thin per-emission signal. So the MsPacman "∼60 % learned / ∼40 % architectural" decomposition is *not* universal: the learned fraction ranges from ∼100 % (RoadRunner) to negative (River Raid), and its sign is predicted by whether $T > F$. This is the mechanistic complement to the $r = 0.92$ predictor: the latent helps where—and only where—it has carried real content from a slow channel that itself beats fast-only.

L 30B-A3B cross-scale ablation (full numbers)

Replicates the MsPacman/bare F/T/L cells with the slow model swapped from Qwen3-VL-8B-Thinking (dense) to Qwen3-VL-30B-A3B-Thinking (MoE, ∼3B active per forward, 48 layers, hidden_dim=2048). Pipeline:

- **Stage B**: 15 T-trajectories collected with the 30B slow (~470 emissions, ~7200 ticks). Same prompt template as 8B.
- **Stage C v2**: re-trained projection 2048 → 4096 (~33M parameters, same 2-layer MLP topology), 1 epoch over ~7K samples. Final mean KL = 0.015 (*vs* 8B's ~0.005; slightly under-converged given ~3× less data per slow-model parameter).
- **Eval**: F/T/L on MsPacman, 6 seeds × 2 episodes per cell, $n = 12$, ≤500 ticks/episode. Run twice: greedy and sample $\tau = 1.0$.

Configuration	F	T	L	Δ_{L-T}
8B-dense, greedy (paper)	256 ± 25	408 ± 92	$\mathbf{628 \pm 356}$	**+54%** (Welch $p = 0.06$)
30B-A3B, greedy, $n = 12$	273 ± 45	312 ± 14	210 ± 0	−33% (det. artifact, Q*bert-style)
30B-A3B, sample $\tau = 1.0$, $n = 12$	309 ± 81	389 ± 209	$\mathbf{458 \pm 283}$	+18 % (Welch $p = 0.50$, $d = 0.28$)

Reading. A "bigger slow model carries more, so $L - T$ should grow with capacity" hypothesis is not supported here. With Qwen3-VL-30B-A3B as slow, the L > T direction is preserved under sampling (+18 %) but the magnitude *shrinks* relative to 8B-dense (+54 %), and significance is lost ($p = 0.50, n = 12$). The result is consistent with several non-exclusive readings:

- **Active parameters, not total.** 30B-A3B activates only ~3B of its 30B parameters per forward pass. A dense-30B comparison (which we cannot run on this GPU) would be the right capacity test.
- **Hidden-dim shift.** 30B-A3B has hidden_dim=2048 vs 8B's 4096, so the trainable projection maps 2048 → 4096 rather than 4096 → 4096. The same parameter budget covers a harder map.
- **Bridge under-training.** We trained on ~470 30B emissions; the 8B paper baseline used ~5K. Final KL 0.015 vs 0.005 is consistent with partial convergence.
- **One-game evidence.** A single game on a single cross-scale data point is not strong enough to falsify the scaling claim; we report this as a single counter-data point, not as a refutation.

The greedy result ($L = 210 \pm 0$) reproduces the deterministic-trajectory pattern we documented for Q*bert under greedy in the main text—another case where greedy decoding interacts pathologically with the bridge's low per-emission variance.

M Per-game collapse triage

Across the 14 (game, variant) cells in Appendix A, five have a strategy cell at zero (the first five rows below). Robust SA additionally destroys one already-working cell without driving it to exactly zero (MsPacman/robust, $L \to 60$), and a separate game, Pong (outside the 14), sits at the random-policy floor (−21). Stage A val-accuracy is the load-bearing predictor of collapse, not bridge-training failure: σ of T, L at deployment correlates with bare Stage A val-acc, not with Stage C KL.

Game/variant	Collapsed cell	Diagnosis
RoadRunner/bare	$F = 0$	Stage A predicts NOOP; bridges rescue.
SpaceInvaders/bare	$T = L = 0$	Stage A OOD-brittle; robust partially rescues.
Q*bert/bare	$T = L = 0$	Stage A OOD-brittle; robust rescues to $T = 125, L = 50$.
Seaquest/robust	$T = L = 0$	Robust drops below bare F; use bare.
Enduro/bare	$T = 0$	Text collapses; robust gives a non-degenerate near-floor tie.
MsPacman/robust	$L \to 60$	(non-zero) Robust destroys MsPacman policy; use bare.
Pong/all	all −21	(outside 14) Stage A val-acc only 1.5× random; cannot be rescued.

N MetaDrive (non-Atari) full numbers

Setup. MetaDrive [13] at ~10 Hz control. The fast model sees a top-down rendered 84×84 image (rendered via the pygame top-down channel, no GL—this is what makes headless end-to-end runs feasible on one GPU); 9 discrete actions (3 steer $\times$ 3 throttle). The slow model sees the structured state (speed, heading, lane offset, upcoming-turn cue derived from `navi_arrow_dir`, nearby vehicles). Stage A clones MetaDrive's built-in PPO [20] expert; Stage B collects slow emissions under expert driving; Stage C distils the latent against the (robust, suffix-aware) text teacher. Random policy scores $\sim$8 reward; the discrete expert scores $\sim$174–211.

Two task regimes. *Reactive* = default 3-block map (lane-keeping dominates). *Planning* = map `SXSXSX` (straights + X-intersections) with off-route termination, so survival requires correct turn decisions (the route expert beats a straight-driving policy by +218 reward). Eval $n=8$ (4 seeds $\times$ 2 episodes) on the robust head.

Regime	Decoder	F	T	L
Reactive	greedy	71.2	69.5	69.5
Planning	greedy	87.8	85.1	85.1
Planning	sample $\tau=1$	123.2	43.5	36.7

Bridge-replacement control (planning, greedy, $n = 8$). $L = 85.1$, $L_{\text{zero}} = 89.5$, $L_{\text{random}} = 97.2$: replacing the trained latent with zeros or matched-norm random vectors does *not* reduce the score (if anything it rises), i.e. the latent is inert—it carries no behaviour-relevant information. This is the opposite of MsPacman ($L_{\text{trained}} = 628 \gg L_{\text{random}} = 387$, §K), and is the diagnostic basis for calling MetaDrive a controlled negative.

Why L cannot exceed F here. Stage C trains L to match the text teacher's action distribution (objective $\mathrm{KL}(\pi_L \| \pi_T)$), so L's ceiling is T. On MetaDrive $T \le F$ in every configuration we ran—both task regimes (reactive lane-keeping, planning intersections), both decoders (greedy, sampling), and both Stage A heads: the bare head, where the text channel collapses ($T=17.2$ vs $F=73.7$), and the robust head, where T sits just below F in the tabulated reactive (69.5 vs 71.2), planning-greedy (85.1 vs 87.8), and planning-sampling (43.5 vs 123.2) cells—so the best attainable latent is $L \approx F$. Fixing the teacher (robust, suffix-aware head, expert-driven data, KL converged to 0.004) makes the distillation *clean*—$L = T$ to the decimal under greedy—but cannot raise a ceiling set by a teacher that does not beat fast-only. The slow model's frame-grounded, $\sim$1.5 s reasoning simply does not produce better driving decisions than the reactive policy, even when the task demands route planning.

Robustness of the predictor. The $(T-F, L-F)$ correlation (Figure 5) is robust to both variant selection and individual games. *Variant selection:* $r = 0.92$ over the 8 best-variant cells and $r = 0.94$ over all 16 (game, variant) cells with no selection—the 16 are the 14 bare/robust Atari cells, a SpaceInvaders expert-data cell (§5), and MetaDrive—so the choice of which Stage A variant to headline does not create the relationship. *Individual games* (on the 8 best-variant cells): leave-one-out Pearson r stays in $[0.89, 0.98]$ for all 8 deletions (dropping the RoadRunner high-leverage point gives $r = 0.89$); the rank correlation is Spearman $\rho = 0.98$; a game-level bootstrap (10,000 resamples) gives a 95 % CI of $[0.81, 1.0]$; and excluding both dynamic-range extremes (RoadRunner and River Raid) still gives $r = 0.84$ ($n = 6$).